INCH-POUND

MIL-DTL-55302/138C
w/AMENDMENT 1
10 September 2018
SUPERSEDING
MIL-DTL-55302/138C
24 March 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, PIN, 160 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

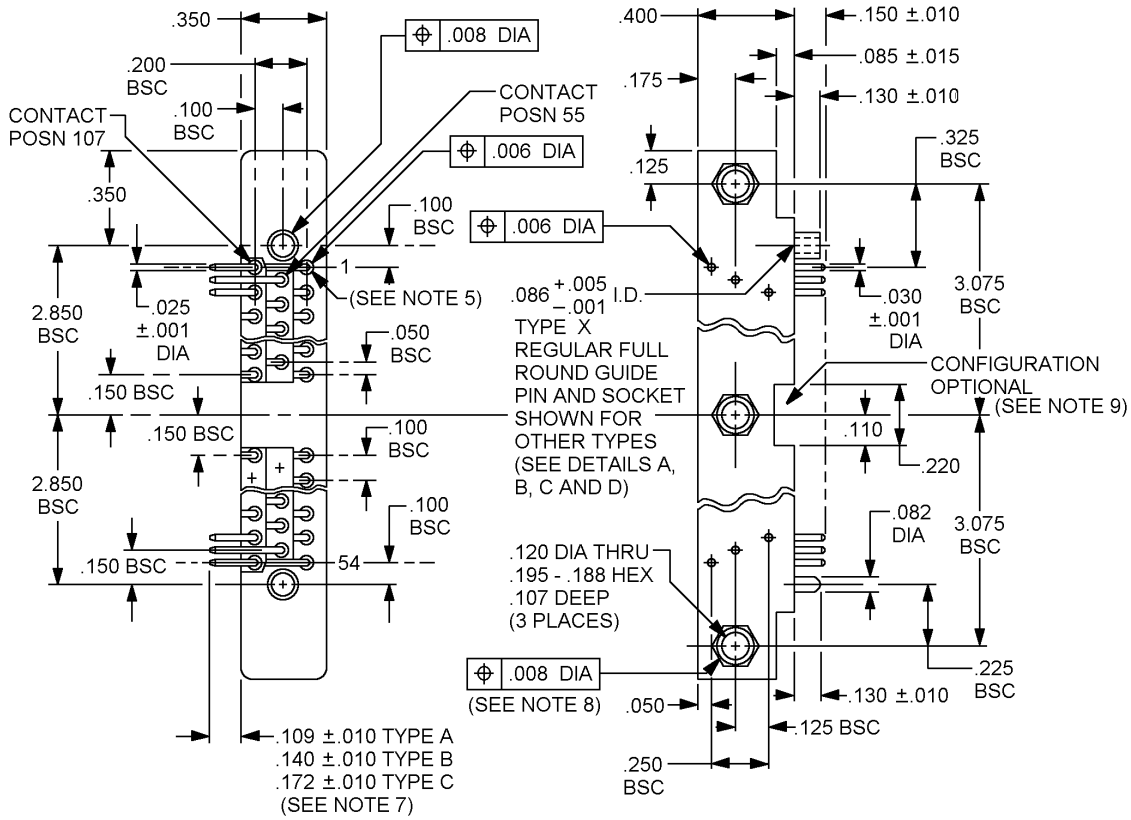
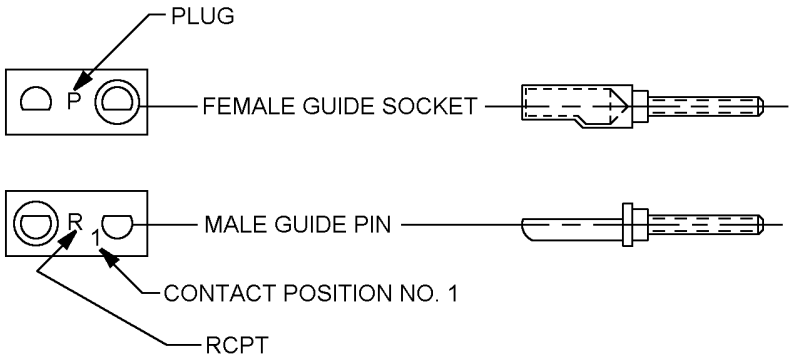


FIGURE 1. Connectors, plug, .100 (25.4 mm) spacing.

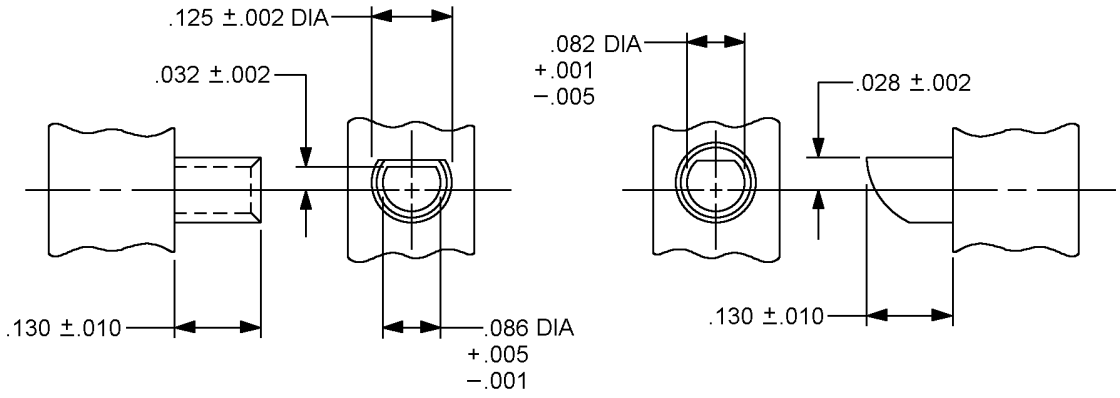
AMSC N/A

FSC 5935

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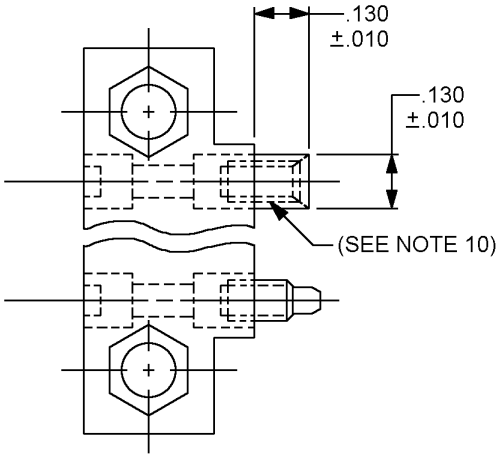
CONNECTOR POLARIZATION WITH "D" SHAPED GUIDE PINS AND SOCKETS. (TYPE "Y")



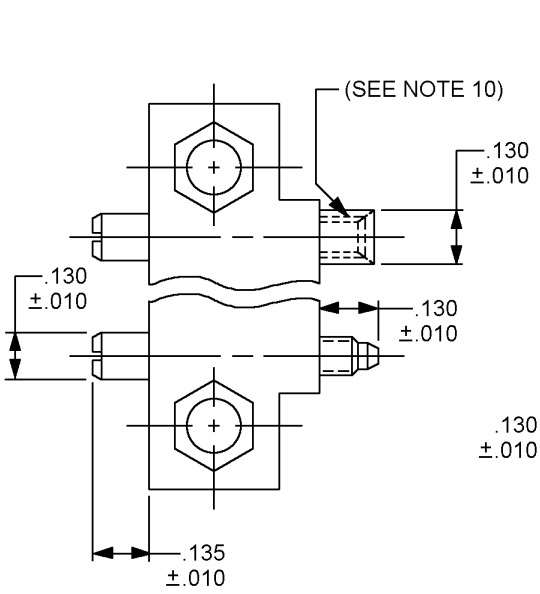
DETAIL "A" POLARIZATION PIN/SOCKET

FIGURE 1. Connectors, plug, .100 (25.4 mm) spacing - Continued.

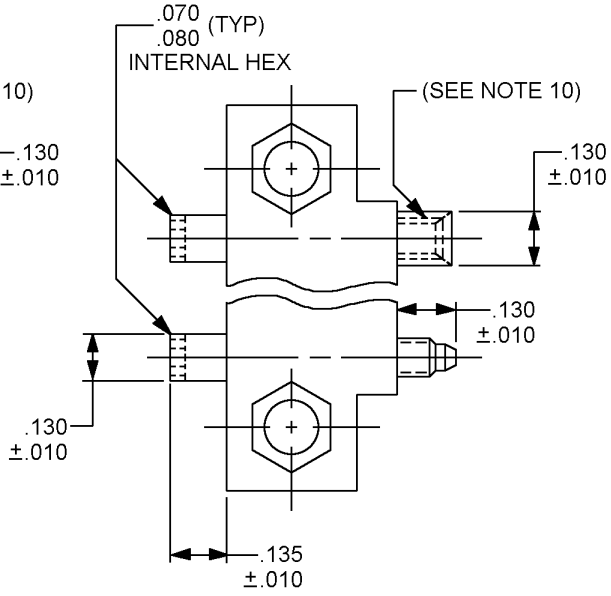
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TYPE "F" FIXED JACKSET
DETAIL "B"



TYPE "S" SHORT TURNING SLOTTED JACKSET
DETAIL "C"



TYPE "H" SHORT TURNING HEX JACKSET
DETAIL "D"

FIGURE 1. Connectors, plug, .100 (25.4 mm) spacing - Continued.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.080	2.03	.172	4.37
.002	0.05	.082	2.08	.175	4.45
.005	0.13	.085	2.16	.188	4.78
.006	0.15	.086	2.18	.195	4.95
.008	0.20	.100	2.54	.200	5.08
.010	0.25	.107	2.72	.220	5.59
.015	0.38	.109	2.77	.225	5.72
.025	0.64	.110	2.79	.250	6.35
.028	0.71	.120	3.05	.325	8.26
.030	0.76	.125	3.18	.350	8.89
.032	0.81	.130	3.30	.400	10.16
.050	1.27	.135	3.43	2.850	72.39
.070	1.78	.140	3.56	3.075	78.11
.078	1.98	.150	3.81	6.400	162.56

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. These connectors mate with connectors specified in MIL-DTL-55302/139.
5. Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the side(s) of the connector. As and option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number one contact shall be marked.
6. Termination layout on .025 (0.64 mm) modular grid.
7. Optional undercut .050 (1.27mm) maximum, on PCB side for cleaning purposes.
8. Position of Hex orientation is optional. Hex shall not break through the side of the insulator.
9. Counter bore or slot.
10. Hole depth to female guide hardware is .282 min (7.16 mm). Full thread depth to female threaded hardware is .240 min (6.1 mm).

FIGURE 1. Connectors, plug, .100 (25.4 mm) spacing - Continued.

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Requirements:

Dimensions and configuration: See figure 1.

Material: Guide pins, type X guide sockets, jackscrews and jack sockets shall be made of corrosion resisting stainless steel in accordance with ASTM A582/A582M passivated in accordance with SAE AMS2700 or shall be corrosion resistant copper nickel alloy (61 ±2% nickel, 2.5% maximum other, and the balance zinc) and shall have a brinell hardness of 145 to 175. Type Y guide sockets shall be beryllium copper in accordance with ASTM B196/B196M or ASTM B197/B197M, nickel plated in accordance with SAE-AMS-QQ-N-290, class 2, 30 to 150 microinches.

Plating: The contact plating shall be in accordance with MIL-DTL-55302 or gold in accordance with MIL-DTL-45204 or equivalent as approved by the qualifying and preparing activities, type II, grade C, class 1, over nickel plating in accordance with SAE-AMS-QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold in accordance with MIL-DTL-45204, type II, grade C, class 00 or type III, grade A, class 00, over nickel plating in accordance with SAE-AMS-QQ-N-290, class 2, 50 to 150 microinches in the termination area.

Nonfunctional areas: See MIL-DTL-55302.

Contact identification: See figure 1.

Current rating: 5.0 amperes maximum in accordance with contact.

Contact resistance: The average contact resistance of all contacts measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:
Sea level: 1,000 volts rms.
High altitude: 300 volts rms.

Polarization: Type X, regular, guide pin and socket, full round (see figure 1). Type Y, adjustable “D” shaped guide pin and/or socket (see detail A and MIL-DTL-55302/57, detail B). All connectors are supplied in the -1 polarized position. For other than Original Equipment Manufacturer (OEM) use, connectors may be repolarized by using M55302/57-01 spanner wrench and M55302/57-02 guide pin and guide socket kit. When guides are removed and reinstalled in another position apply Loctite 83-CV or equivalent to the threads. Ordering connectors in other than the -1 polarized position may be done by the OEM’s by placing the desired polarized in brackets three spaces after the Part or Identifying Number (PIN) on the purchase order. Example M55302/138-A160Y [-32]. Polarization pins and sockets shall be adjusted to any desired configuration by spanner wrench (M55302/57-01). Ordering connectors in the -1 polarized position without Loctite 83-CV or equivalent applied may be done by placing a “-0” in brackets three spaces after the part on the purchase order.

M55302/57-02 kit: Consists of one type Y “D” shaped guide socket and spanner nut and one type Y “D” shaped guide pin and spanner nut.

NOTE: Hardware with Loctite 83-CV or equivalent applied, shall withstand a torque of 7-inch ounces applied through the nut.

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Example of PIN:

M55302/138 A 160 X

Basic number of specification sheet

Type of terminal (see figure 1)

A = Dip terminal .109 inch (2.77 mm) length
B = Dip terminal .140 inch (3.56 mm) length
C = Dip terminal .172 inch (4.37 mm) length

Number of contacts (see figure 1)

Type mounting hardware (see figure 1, details A, B, C, and D)

X = Full round
Y = D-shaped
F = Fixed jackset
S = 135 slotted turning jackset
H = 135 hex turning jackset

Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-45204
MIL-DTL-55302/57
MIL-DTL-55302/139
ASTM A582/A582M
ASTM B196/B196M
ASTM B197/B197M
SAE-AMS-QQ-N-290
SAE AMS2700

CONCLUDING MATERIAL

Custodians:
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2018-115)

Review activities:
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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